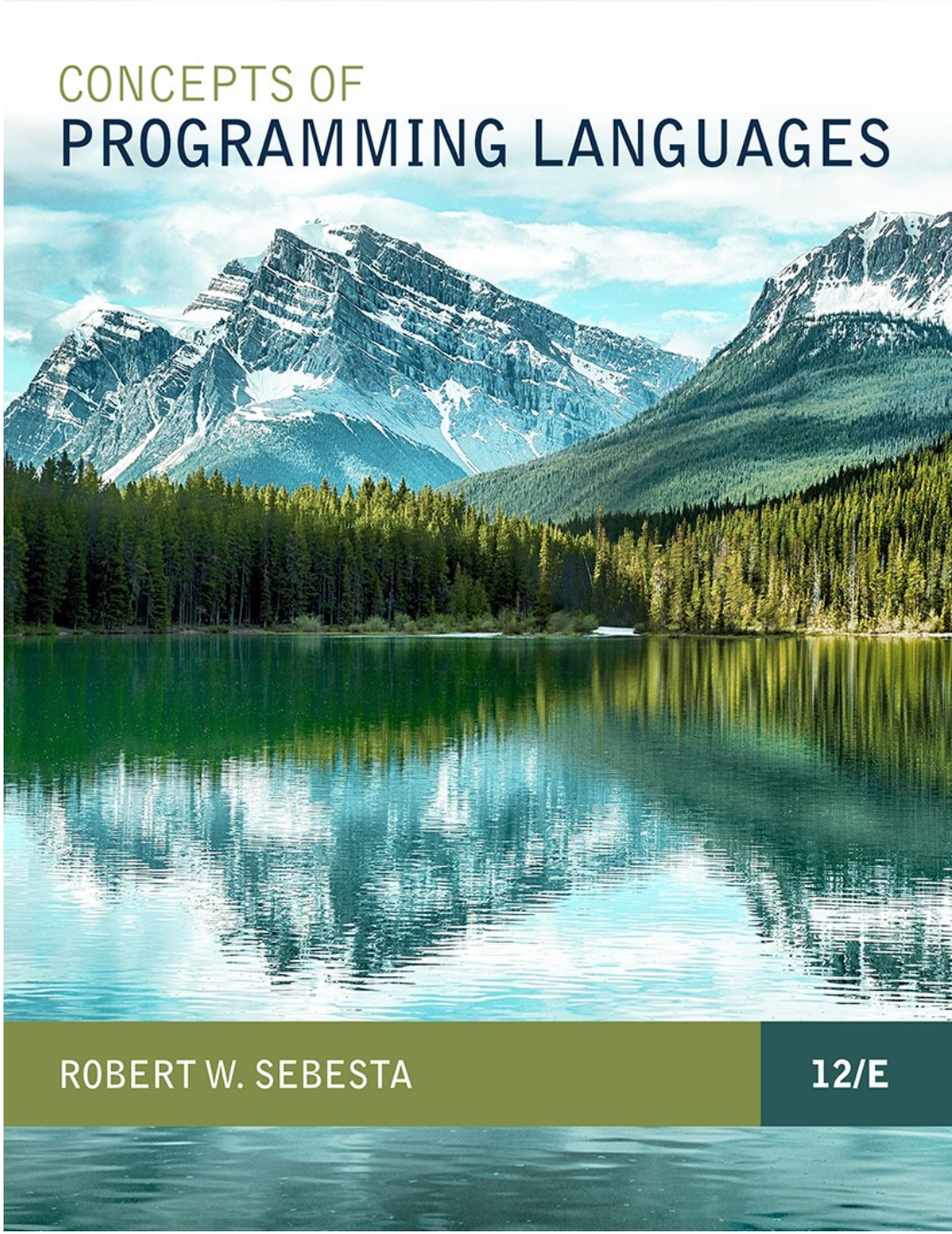
Chapter 3

Describing Syntax and Semantics



ISBN 0-321-49362-1

Chapter 3 Topics

- Introduction
- The General Problem of Describing Syntax
- Formal Methods of Describing Syntax
- Attribute Grammars
- Describing the Meanings of Programs: Dynamic Semantics

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Introduction

- **Syntax**: the form or structure of the expressions, statements, and program units
- **Semantics**: the meaning of the expressions, statements, and program units
- Syntax and semantics provide a language's definition
  - Users of a language definition
    - Other language designers
    - Implementers
    - Programmers (the users of the language)

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The General Problem of Describing Syntax: Terminology

- A *sentence* is a string of characters over some alphabet
- A *language* is a set of sentences
- A *lexeme* is the lowest level syntactic unit of a language (e.g., `*`, `sum`, `begin`)
- A *token* is a category of lexemes (e.g., identifier)

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Formal Definition of Languages

- **Recognizers**
  - A recognition device reads input strings over the alphabet of the language and decides whether the input strings belong to the language
  - Example: syntax analysis part of a compiler
    - Detailed discussion of syntax analysis appears in Chapter 4

- **Generators**
  - A device that generates sentences of a language
  - One can determine if the syntax of a particular sentence is syntactically correct by comparing it to the structure of the generator

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BNF and Context-Free Grammars

- Context-Free Grammars
  - Developed by Noam Chomsky in the mid-1950s
  - Language generators, meant to describe the syntax of natural languages
  - Define a class of languages called context-free languages

- Backus-Naur Form (1959)
  - Invented by John Backus to describe the syntax of Algol 58
  - BNF is equivalent to context-free grammars

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BNF Fundamentals

- In BNF, abstractions are used to represent classes of syntactic structures--they act like syntactic variables (also called *nonterminal symbols,* or just *terminals*)
- *Terminals* are lexemes or tokens
- A rule has a left-hand side (LHS), which is a nonterminal, and a right-hand side (RHS), which is a string of terminals and/or nonterminals

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BNF Fundamentals (continued)

- Nonterminals are often enclosed in angle brackets
  - Examples of BNF rules:

```
<ident_list> → identifier | identifier, <ident_list>
<if_stmt> → if <logic_expr> then <stmt>
```

- Grammar: a finite non-empty set of rules
- A *start symbol* is a special element of the nonterminals of a grammar

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BNF Rules

- An abstraction (or nonterminal symbol) can have more than one RHS

```
<stmt> → <single_stmt>
       | begin <stmt_list> end
```

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Describing Lists

- Syntactic lists are described using recursion

```
<ident_list> → ident
               | ident, <ident_list>
```

- A derivation is a repeated application of rules, starting with the start symbol and ending with a sentence (all terminal symbols)

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An Example Grammar

```
<program> → <stmts>
 <stmts> → <stmt> | <stmt> ; <stmts>
 <stmt> → <var> = <expr>
 <var> → a | b | c | d
 <expr> → <term> + <term> | <term> - <term>
 <term> → <var> | const
```

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An Example Derivation

```
<program> => <stmts> => <stmt>
                     => <var> = <expr>
                     => a = <expr>
                     => a = <term> + <term>
                     => a = <var> + <term>
                     => a = b + <term>
                     => a = b + const
```

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Derivations

- Every string of symbols in a derivation is a *sentential form*
- A *sentence* is a sentential form that has only terminal symbols
- A *leftmost derivation* is one in which the leftmost nonterminal in each sentential form is the one that is expanded
- A derivation may be neither leftmost nor rightmost

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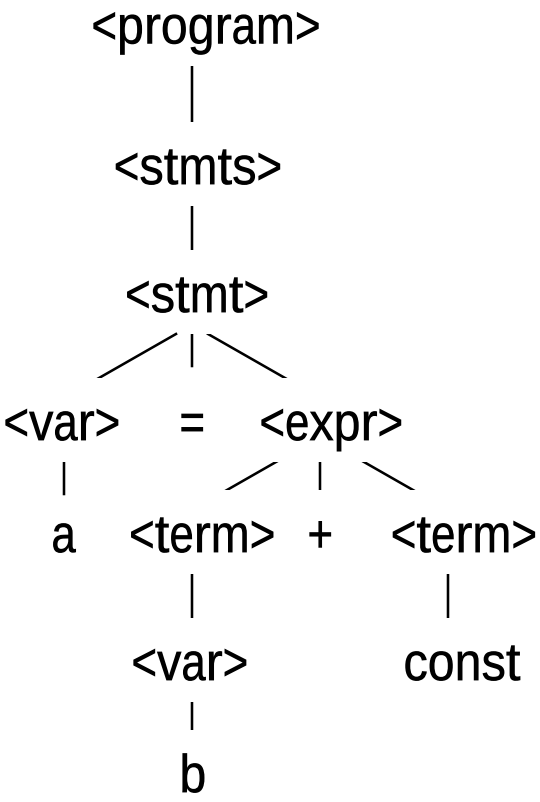
Parse Tree

- A hierarchical representation of a derivation



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Ambiguity in Grammars

- A grammar is *ambiguous* if and only if it generates a sentential form that has two or more distinct parse trees

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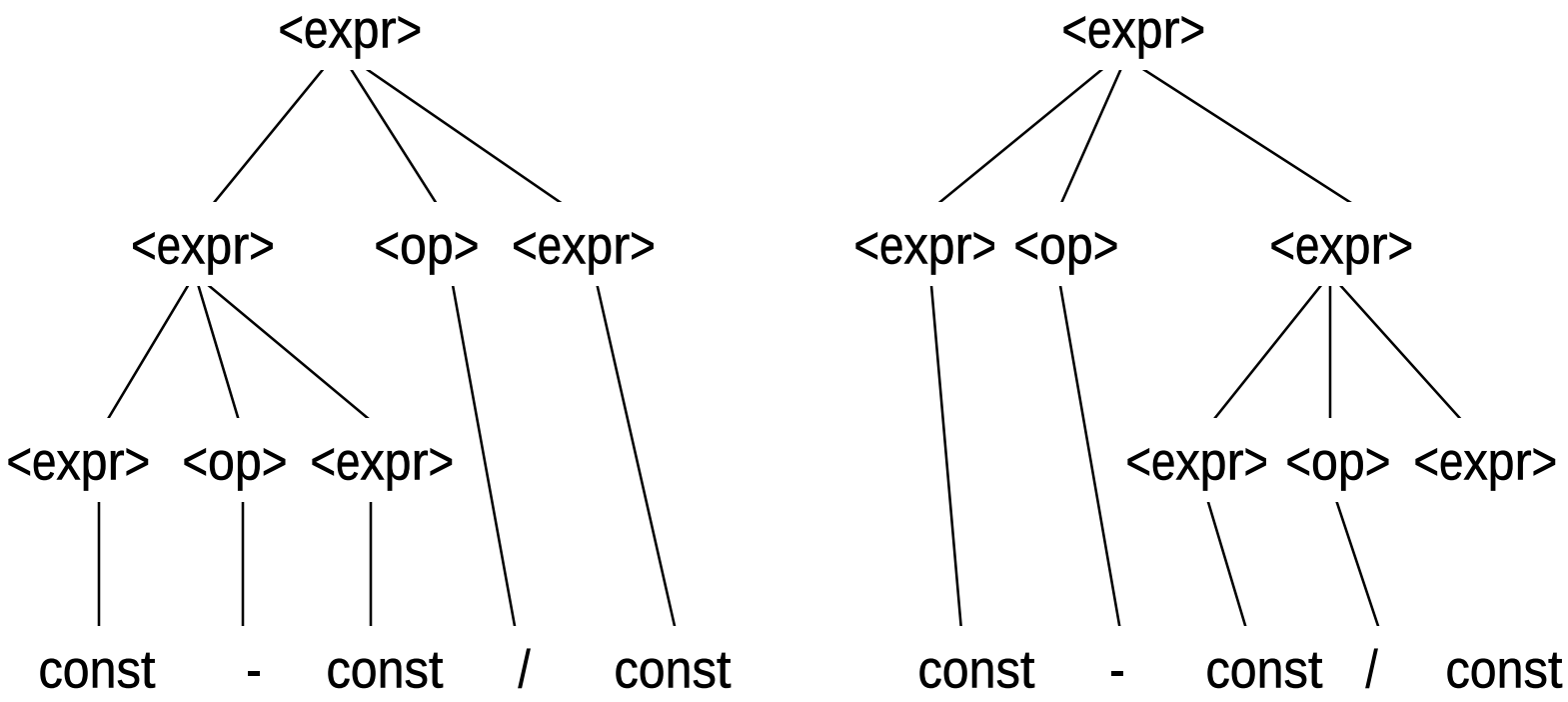
An Ambiguous Expression Grammar

<expr> → <expr> <op> <expr> | const
<op> → / | -



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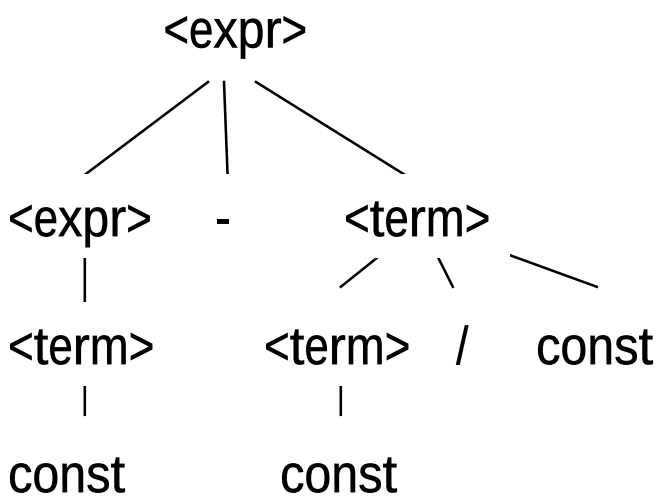
An Unambiguous Expression Grammar

- If we use the parse tree to indicate precedence levels of the operators, we cannot have ambiguity

```
<expr> → <expr> - <term>  |  <term>
<term> → <term> / const| const
```



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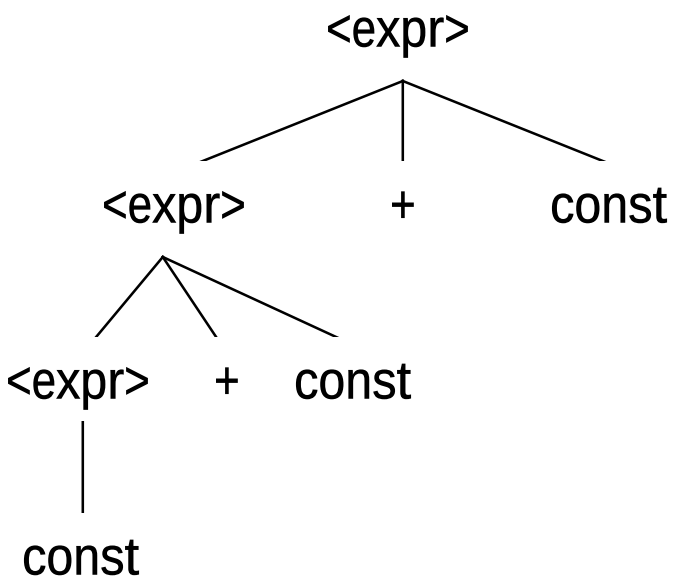
Associativity of Operators

- Operator associativity can also be indicated by a grammar

```
<expr> -> <expr> + <expr> |  const  (ambiguous)
<expr> -> <expr> + const  |  const  (unambiguous)
```



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Extended BNF

- Optional parts are placed in brackets [ ]

  `<proc_call> -> ident [(<expr_list>)]`

- Alternative parts of RHSs are placed inside parentheses and separated via vertical bars

  `<term> → <term> (+|-) const`

- Repetitions (0 or more) are placed inside braces { }

  `<ident> → letter {letter|digit}`

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BNF and EBNF

- BNF

```
<expr> → <expr> + <term>
       | <expr> - <term>
       | <term>
<term> → <term> * <factor>
       | <term> / <factor>
       | <factor>
```

- EBNF

```
<expr> → <term> {(+ | -) <term>}
<term> → <factor> {(* | /) <factor>}
```

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Recent Variations in EBNF

- Alternative RHSs are put on separate lines
- Use of a colon instead of `=>`
- Use of `opt` for optional parts
- Use of `oneof` for choices

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Static Semantics

- Nothing to do with meaning
- Context-free grammars (CFGs) cannot describe all of the syntax of programming languages
- Categories of constructs that are trouble:
  - Context-free, but cumbersome (e.g., types of operands in expressions)
  - Non-context-free (e.g., variables must be declared before they are used)

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Attribute Grammars

- Attribute grammars (AGs) have additions to CFGs to carry some semantic info on parse tree nodes

- Primary value of AGs:
  - Static semantics specification
  - Compiler design (static semantics checking)

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Attribute Grammars : Definition

- Def: An attribute grammar is a context-free grammar G = (S, N, T, P) with the following additions:
  - For each grammar symbol *x* there is a set *A(x)* of attribute values
  - Each rule has a set of functions that define certain attributes of the nonterminals in the rule
  - Each rule has a (possibly empty) set of predicates to check for attribute consistency

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Attribute Grammars: Definition

- Let $X_0 \rightarrow X_1 \ldots X_n$ be a rule
- Functions of the form $S(X_0) = f(A(X_1), \ldots, A(X_n))$ define *synthesized attributes*
- Functions of the form $I(X_j) = f(A(X_0), \ldots, A(X_n))$, *for* $i <= j <= n$, define *inherited attributes*
- Initially, there are *intrinsic attributes* on the leaves

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Attribute Grammars: An Example

- Syntax

  `<assign> -> <var> = <expr>`

  `<expr> -> <var> + <var> | <var>`

  `<var> A | B | C`

- `actual_type`: synthesized for `<var>` and `<expr>`
- `expected_type`: inherited for `<expr>`

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Attribute Grammar (continued)

- Syntax rule: `<expr>` → `<var>[1] + <var>[2]`

  Semantic rules:

  `<expr>.actual_type` ← `<var>[1].actual_type`

  Predicate:

  `<var>[1].actual_type == <var>[2].actual_type`

  `<expr>.expected_type == <expr>.actual_type`

- Syntax rule: `<var>` → `id`

  Semantic rule:

  `<var>.actual_type` ← `lookup (<var>.string)`

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Attribute Grammars (continued)

- How are attribute values computed?
  - If all attributes were inherited, the tree could be decorated in top-down order.
  - If all attributes were synthesized, the tree could be decorated in bottom-up order.
  - In many cases, both kinds of attributes are used, and it is some combination of top-down and bottom-up that must be used.

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Attribute Grammars (continued)

```
<expr>.expected_type ← inherited from parent

<var>[1].actual_type ← lookup (A)
<var>[2].actual_type ← lookup (B)
<var>[1].actual_type =? <var>[2].actual_type

<expr>.actual_type ← <var>[1].actual_type
<expr>.actual_type =? <expr>.expected_type
```

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Semantics

- There is no single widely acceptable notation or formalism for describing semantics
- Several needs for a methodology and notation for semantics:
  - Programmers need to know what statements mean
  - Compiler writers must know exactly what language constructs do
  - Correctness proofs would be possible
  - Compiler generators would be possible
  - Designers could detect ambiguities and inconsistencies

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Operational Semantics

- Operational Semantics
  - Describe the meaning of a program by executing its statements on a machine, either simulated or actual. The change in the state of the machine (memory, registers, etc.) defines the meaning of the statement
- To use operational semantics for a high-level language, a virtual machine is needed

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Operational Semantics

- A *hardware* pure interpreter would be too expensive
- A *software* pure interpreter also has problems
  - The detailed characteristics of the particular computer would make actions difficult to understand
  - Such a semantic definition would be machine-dependent

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Operational Semantics (continued)

- A better alternative: A complete computer simulation
- The process:
  - Build a translator (translates source code to the machine code of an idealized computer)
  - Build a simulator for the idealized computer
- Evaluation of operational semantics:
  - Good if used informally (language manuals, etc.)
  - Extremely complex if used formally (e.g., VDL), it was used for describing semantics of PL/I.

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Operational Semantics (continued)

- Uses of operational semantics:
  - Language manuals and textbooks
  - Teaching programming languages

- Two different levels of uses of operational semantics:
  - Natural operational semantics
  - Structural operational semantics

- Evaluation
  - Good if used informally (language manuals, etc.)
  - Extremely complex if used formally (e.g.,VDL)

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Denotational Semantics

- Based on recursive function theory
- The most abstract semantics description method
- Originally developed by Scott and Strachey (1970)

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Denotational Semantics – continued

- The process of building a denotational specification for a language:
  - Define a mathematical object for each language entity
  - Define a function that maps instances of the language entities onto instances of the corresponding mathematical objects
- The meaning of language constructs are defined by only the values of the program's variables

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Denotational Semantics: program state

- The state of a program is the values of all its current variables

$$s = \{<i_1, v_1>, <i_2, v_2>, \ldots, <i_n, v_n>\}$$

- Let **VARMAP** be a function that, when given a variable name and a state, returns the current value of the variable

$$\text{VARMAP}(i_j, s) = v_j$$

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Decimal Numbers

```
<dec_num> →  '0' | '1' | '2' | '3' | '4' | '5' |
             '6' | '7' | '8' | '9' |
             <dec_num> ('0' | '1' | '2' | '3' |
                        '4' | '5' | '6' | '7' |
                        '8' | '9')
```

M_{dec}('0') = 0, M_{dec} ('1') = 1, …, M_{dec} ('9') = 9

M_{dec} (<dec_num> '0') = 10 * M_{dec} (<dec_num>)

M_{dec} (<dec_num> '1') = 10 * M_{dec} (<dec_num>) + 1

…

M_{dec} (<dec_num> '9') = 10 * M_{dec} (<dec_num>) + 9

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Expressions

- Map expressions onto Z $\cup$ {error}
- We assume expressions are decimal numbers, variables, or binary expressions having one arithmetic operator and two operands, each of which can be an expression

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Expressions

```
Me(<expr>, s) Δ=
    case <expr> of
      <dec_num> => Mdec(<dec_num>, s)
      <var> =>
           if VARMAP(<var>, s) == undef
                then error
                else VARMAP(<var>, s)
     <binary_expr> =>
          if (Me(<binary_expr>.<left_expr>, s) == undef
                OR Me(<binary_expr>.<right_expr>, s) =
                                 undef)
               then error
           else
           if (<binary_expr>.<operator> == '+' then
              Me(<binary_expr>.<left_expr>, s) +
                     Me(<binary_expr>.<right_expr>, s)
           else Me(<binary_expr>.<left_expr>, s) *
               Me(<binary_expr>.<right_expr>, s)
        ...
```

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Assignment Statements

- Maps state sets to state sets U {error}

```
Ma(x := E, s) Δ=
    if Me(E, s) == error
       then error
       else s' =
      {<i1,v1'>,<i2,v2'>,...,<in,vn'>},
              where for j = 1, 2, ..., n,
                  if ij == x
                    then vj' = Me(E, s)
                    else vj' = VARMAP(ij, s)
```

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Logical Pretest Loops

- Maps state sets to state sets U {error}

```
Ml(while B do L, s) Δ=
    if Mb(B, s) == undef
        then error
        else if Mb(B, s) == false
            then s
            else if Msl(L, s) == error
                then error
                else Ml(while B do L, Msl(L, s))
```

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Loop Meaning

- The meaning of the loop is the value of the program variables after the statements in the loop have been executed the prescribed number of times, assuming there have been no errors
- In essence, the loop has been converted from iteration to recursion, where the recursive control is mathematically defined by other recursive state mapping functions
  - Recursion, when compared to iteration, is easier to describe with mathematical rigor

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Evaluation of Denotational Semantics

- Can be used to prove the correctness of programs
- Provides a rigorous way to think about programs
- Can be an aid to language design
- Has been used in compiler generation systems
- Because of its complexity, it are of little use to language users

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Axiomatic Semantics

- Based on formal logic (predicate calculus)
- Original purpose: formal program verification
- Axioms or inference rules are defined for each statement type in the language (to allow transformations of logic expressions into more formal logic expressions)
- The logic expressions are called *assertions*

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Axiomatic Semantics (continued)

- An assertion before a statement (a *precondition*) states the relationships and constraints among variables that are true at that point in execution
- An assertion following a statement is a *postcondition*
- A *weakest precondition* is the least restrictive precondition that will guarantee the postcondition

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Axiomatic Semantics Form

- Pre-, post form: `{P} statement {Q}`

- An example
  - `a = b + 1  {a > 1}`
  - One possible precondition: `{b > 10}`
  - Weakest precondition: `{b > 0}`

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Program Proof Process

- The postcondition for the entire program is the desired result
  - Work back through the program to the first statement. If the precondition on the first statement is the same as the program specification, the program is correct.

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Axiomatic Semantics: Assignment

- An axiom for assignment statements (x = E): $\{Q_{x->E}\}$ x = E $\{Q\}$

- The Rule of Consequence:

$$\frac{\{P\}\,S\,\{Q\},\ P' \Rightarrow P,\ Q \Rightarrow Q'}{\{P'\}\,S\,\{Q'\}}$$

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Axiomatic Semantics: Sequences

- An inference rule for sequences of the form S1; S2

  {P1} S1 {P2}

  {P2} S2 {P3}

$$\frac{\{P1\}\,S1\,\{P2\},\ \{P2\}\,S2\,\{P3\}}{\{P1\}\,S1;S2\,\{P3\}}$$

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Axiomatic Semantics: Selection

- An inference rules for selection
  - **`if`** B **`then`** S1 **`else`** S2

$$\frac{\{B \text{ and } P\}\ S1\ \{Q\},\ \{(\text{not } B) \text{ and } P\}\ S2\ \{Q\}}{\{P\}\ \textbf{if}\ B\ \textbf{then}\ S1\ \textbf{else}\ S2\ \{Q\}}$$

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Axiomatic Semantics: Loops

- An inference rule for logical pretest loops

{P} **`while`** B **`do`** S **`end`** {Q}

$$\frac{(\text{I and B})\ \text{S}\ \{\text{I}\}}{\{\text{I}\}\ \text{while B do S}\ \{\text{I and (not B)}\}}$$

where I is the loop invariant (the inductive hypothesis)

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Axiomatic Semantics: Axioms

- Characteristics of the loop invariant: I must meet the following conditions:
  - P => I -- the loop invariant must be true initially
  - {I} B {I} -- evaluation of the Boolean must not change the validity of I
  - {I and B} S {I} -- I is not changed by executing the body of the loop
  - (I and (not B)) => Q -- if I is true and B is false, Q is implied
  - The loop terminates -- can be difficult to prove

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Loop Invariant

- The loop invariant I is a weakened version of the loop postcondition, and it is also a precondition.
- I must be weak enough to be satisfied prior to the beginning of the loop, but when combined with the loop exit condition, it must be strong enough to force the truth of the postcondition

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Evaluation of Axiomatic Semantics

- Developing axioms or inference rules for all of the statements in a language is difficult
- It is a good tool for correctness proofs, and an excellent framework for reasoning about programs, but it is not as useful for language users and compiler writers
- Its usefulness in describing the meaning of a programming language is limited for language users or compiler writers

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Denotation Semantics vs Operational Semantics

- In operational semantics, the state changes are defined by coded algorithms
- In denotational semantics, the state changes are defined by rigorous mathematical functions

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Summary

- BNF and context-free grammars are equivalent meta-languages
  - Well-suited for describing the syntax of programming languages
- An attribute grammar is a descriptive formalism that can describe both the syntax and the semantics of a language
- Three primary methods of semantics description
  - Operation, axiomatic, denotational

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